

SONiX 32-Bit Cortex-M0 Micro-Controller

UART Example Code

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AMENDMENT HISTORY

Version	Date	Description
1.0	2024/12/02	1. First version.

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1 OVERVIEW

This document provides our customers the C program examples of UART.

2 FEATURE

- Interrupt access support transmitter & receiver mode.
- Use ARM-compiler V6.12

3 HW

- The starter kit board of vendor desired MCU.
- Connect UTXD0 and URXD0.

4 EXAMPLE CODE

1. Use UART to transmit the data from 0x61 to 0x68.

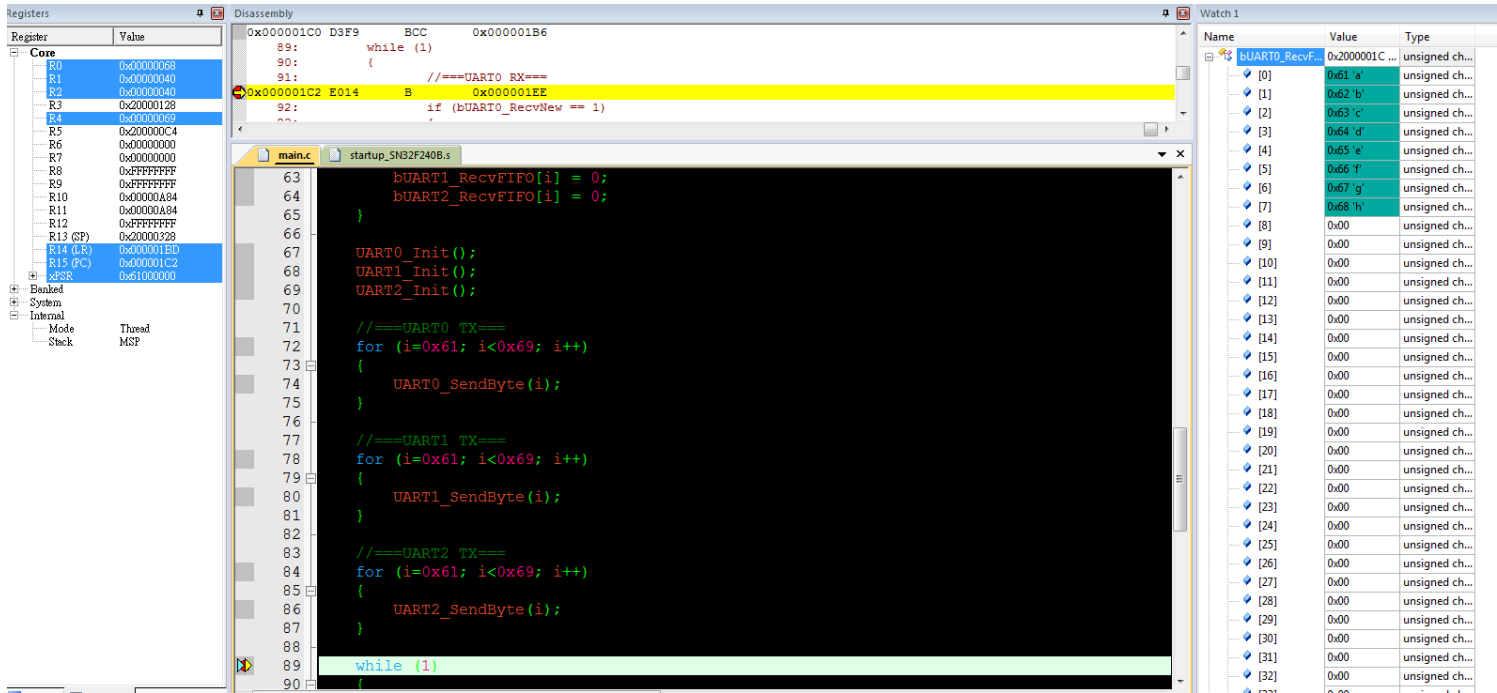
```
//===UART0 TX===  
for (i=0x61; i<0x69; i++)  
{  
    UART0_SendByte(i);  
}
```

2. Call USART0_Init(); or UART0_Init();

```
int main (void)  
{  
    //init USART0  
    USART0_Init();  
  
    .  
    .  
    User code  
    .  
    .  
}
```

5 RESULT

1. The program will stop at the breakpoint in the main loop



The screenshot shows a debugger interface with the following components:

- Registers:** A list of CPU registers (R0-R15, SP, LR, PC) and their current values. R0 is 0x00000068, R1 is 0x00000040, R2 is 0x00000040, R3 is 0x00000128, R4 is 0x00000128, R5 is 0x000000C4, R6 is 0x00000000, R7 is 0x00000000, R8 is 0xFFFFFFFF, R9 is 0xFFFFFFFF, R10 is 0x00000A84, R11 is 0x00000A84, R12 is 0xFFFFFFFF, R13 (SP) is 0x20000328, R14 (LR) is 0x000001ED, R15 (PC) is 0x000001C2, and PC is 0x000001C2.
- Disassembly:** The main window shows the disassembly of the code. A breakpoint is set at line 91, which is highlighted in yellow. The code is as follows:


```

89:      while (1)
90:      {
91:          //===UART0 RX===
0x000001C2 E014 B 0x000001EE
92:          if (bUART0_RecvNew == 1)
          {
          }
      }
      
```
- main.c:** The source code file is open, showing the main loop. The code is as follows:


```

63      bUART1_RecvFIFO[i] = 0;
64      bUART2_RecvFIFO[i] = 0;
65      }
66
67      UART0_Init();
68      UART1_Init();
69      UART2_Init();
70
71      //===UART0 TX===
72      for (i=0x61; i<0x69; i++)
73      {
74          UART0_SendByte(i);
75      }
76
77      //===UART1 TX===
78      for (i=0x61; i<0x69; i++)
79      {
80          UART1_SendByte(i);
81      }
82
83      //===UART2 TX===
84      for (i=0x61; i<0x69; i++)
85      {
86          UART2_SendByte(i);
87      }
88
89      while (1)
90      {
      
```
- Watch 1:** A table showing the values of the UART0_RecvFIFO registers. The values are 0x61 through 0x68, which correspond to the ASCII characters 'a' through 'h'. The type is 'unsigned char'.

2. UART transmits data from 0x61 to 0x68 as below.



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Main Office:

Address: 10F-1, NO. 36, Taiyuan Street., Chupei City, Hsinchu, Taiwan R.O.C.

Tel: 886-3-5600 888

Fax: 886-3-5600 889

Taipei Office:

Address: 15F-2, NO. 171, Song Ted Road, Taipei, Taiwan R.O.C.

Tel: 886-2-2759 1980

Fax: 886-2-2759 8180

Hong Kong Office:

Unit No.705,Level 7 Tower 1,Grand Central Plaza 138 Shatin Rural Committee Road, Shatin, New Territories, Hong Kong.

Tel: 852-2723-8086

Fax: 852-2723-9179

Technical Support by Email:

Sn8fae@sonix.com.tw